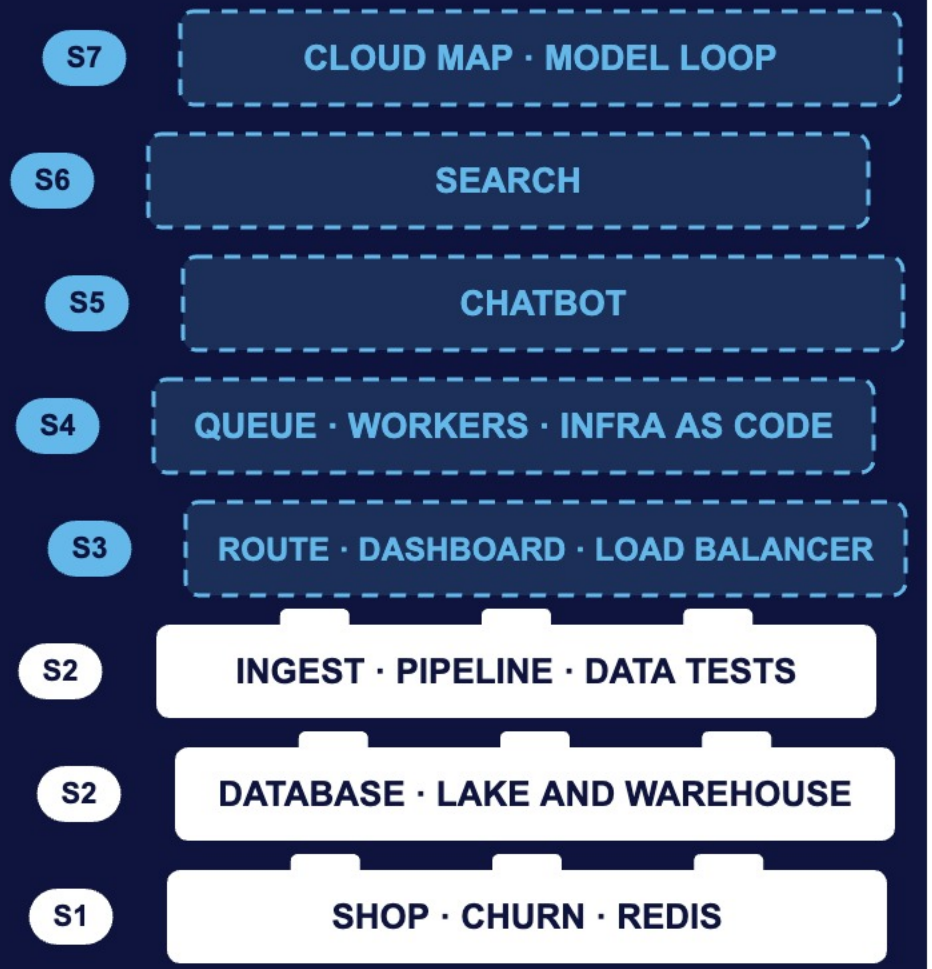


ADVANCED CLOUD AI INFRASTRUCTURE

SESSION 2 · PROJECT KICKOFF

"YOU PICK THE PROBLEM."

Your project: fix one Popote problem, and prove it pays

The course hands you the bricks, one more every session.
Groups of four, from tonight to the oral on 27 November.

Thomas Blasi
t.blasi27@gmail.com

MSA-DAT09-05 · Albert School · Monday 9 November 2026

MARGAUX FUNDS THE IDEAS THAT CAN PROVE THEY PAY

KICKOFF

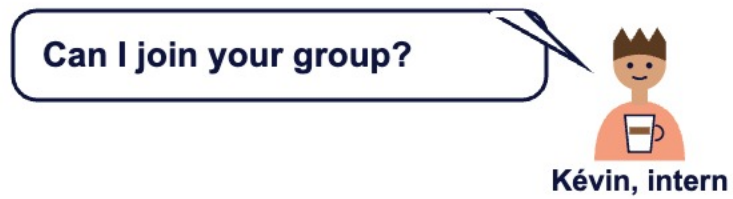
The game: each group pitches one product to Popote's CEO.

POPOTE INVESTMENT COMMITTEE
FRI 27 NOV, FROM 09:00



Margaux, CEO

1	"Whose problem is it, and what does it cost us today?"	ANSWERED BY THE BUSINESS CASE
2	"What does your product change, and how will we know?"	ANSWERED BY THE ARCHITECTURE
3	"What breaks when we're ten times bigger, and who fixes it at 19:05 on a Friday?"	ANSWERED BY THE OPERATIONS



Kévin, intern

Everything you write must answer one of these three questions.

IF YOUR PLAN IS SOLID, THE AI WRITES THE CODE

IN WORDS

What matters in your project, and what doesn't.

WHAT MATTERS	WHAT DOESN'T
The problem, and the KPIs that prove it's solved	How much code you wrote
The architecture, and why each box is there	Whether you or an AI wrote it
What each box can break, and what you decided	How many services you have
How you see it break, and what you do then	
Your answers at the oral, each of you alone	

Use an AI assistant to write the code: it's expected. If it wrote a line you can't explain, it shows at the oral.

A small product you can explain beats a big one you can't.

PART 1 OF 3

One idea, end to end

What a good project looks like

The win-back engine walked through the seven steps, then five checks for your own idea.

1 · ONE IDEA, END TO END

2 · IDEA CARDS

3 · HAND IN, ORAL, CALENDAR

ONE IDEA, SEVEN STEPS: THE WIN-BACK ENGINE

Idea card A walked through the seven steps, so you see the finished shape.

IDEA A · THE WIN-BACK ENGINE

Spot who is about to leave, and win them back for less than they're worth.

STEP 1 DECISION	STEP 2 VALUE	STEP 3 FUNCTIONAL VIEW	STEP 4 NUMBERS	STEP 5 BOXES	STEP 6 TRADE-OFF	STEP 7 RUN IT
Which business decision does your product change, and who makes it?	Which euro line moves? What if you're wrong by half?	What happens, step by step, in business words, no technology names?	How fast, how fresh, how available, how much per 1,000, whose personal data?	Which box does each step? Where does each piece of data live, and how does it move?	Two to four designs, their price, the one you chose, what would make you change	What you watch, the three likeliest failures, what ages and who refreshes it
BUSINESS.md		ARCHITECTURE.md			DECISIONS.md	OPERATIONS.md

Four dials price every design: speed, freshness, cost, complexity.

The same seven steps in every curveball and in your project: each step lands in one file.

STEPS 1 AND 2: THE DECISION IT CHANGES, AND THE KPIS THAT PROVE IT



BASICS

The top of BUSINESS.md for the win-back engine.

1 DECISION · 2 VALUE · 3 · 4 · 5 · 6 · 7

THE PAIN	**About 15 customers in 100 stop ordering each month. Each one is about 100 € of future margin.** (illustrative)
THE DECISION THAT CHANGES	**Who gets a coupon, and how much. Today: nobody, or everyone.** Who decides: marketing, every morning.
SUCCESS CRITERIA	**Customers still ordering 30 days later, targeted vs a held-out group that got nothing. Coupon spend per customer kept, below what a kept customer is worth.**
THE GUARDRAIL	**Coupons used by customers who would have stayed.**
THE VALUE LINE	**leavers brought back × 100 € − coupons used × 8 €**
WRONG BY HALF	**If the coupon brings back half as many, does it still pay?**

STEP 3: WRITE WHAT HAPPENS, IN BUSINESS WORDS, BEFORE ANY BOX

BASICS

The functional view: the first page of ARCHITECTURE.md. No technology names.

1 2 **3 FUNCTIONAL VIEW** 4 5 6 7

STEP	WHO STARTS IT	DATA IT NEEDS	HOW FRESH
Find who is about to leave	every night	orders, deliveries, complaints	yesterday
Pick who gets what	a rule, a budget	the risk, the customer’s value	yesterday
Send the offer	the system, at 9:00	the offer, her phone	now
She orders, or not	the customer	the coupon code	within 30 days
Did it pay?	Margaux, monthly	who got it, who came back, a group that got nothing	after 30 days

THE LAST ROW IS THE HELD-OUT GROUP: THE A/B TEST (SESSION 3)

If you can’t fill this table, no AI will understand your product for you.

STEP 4: FIVE NUMBERS DECIDE WHICH BOXES YOU NEED



BASICS

For the win-back engine, each requirement as a number.

1 2 3 **4 NUMBERS** 5 6 7

THE NUMBER	THE QUESTION	FOR THE WIN-BACK ENGINE
Speed	Who waits, and how long?	**Nobody waits: minutes are fine**
Freshness	How old may the data be?	**Yesterday's orders**
Availability	What if it's down?	**A failed night means coupons a day late**
Cost	How much, per 1,000 customers?	**A small nightly job, plus the coupons**
Personal data	Whose, and which?	**Names and phones: GDPR**

Relaxed numbers mean simple boxes. If marketing said "within a minute", every box would change.

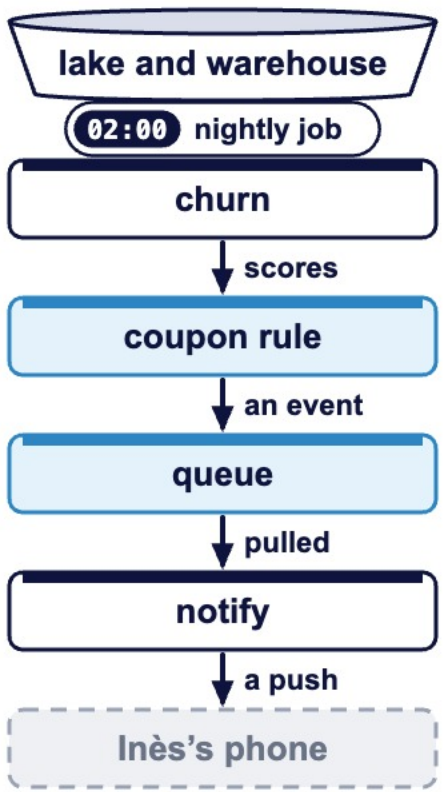
STEP 5: TWO DRAWINGS, THE SYSTEM VIEW AND THE DETAILED ARCHITECTURE

BASICS

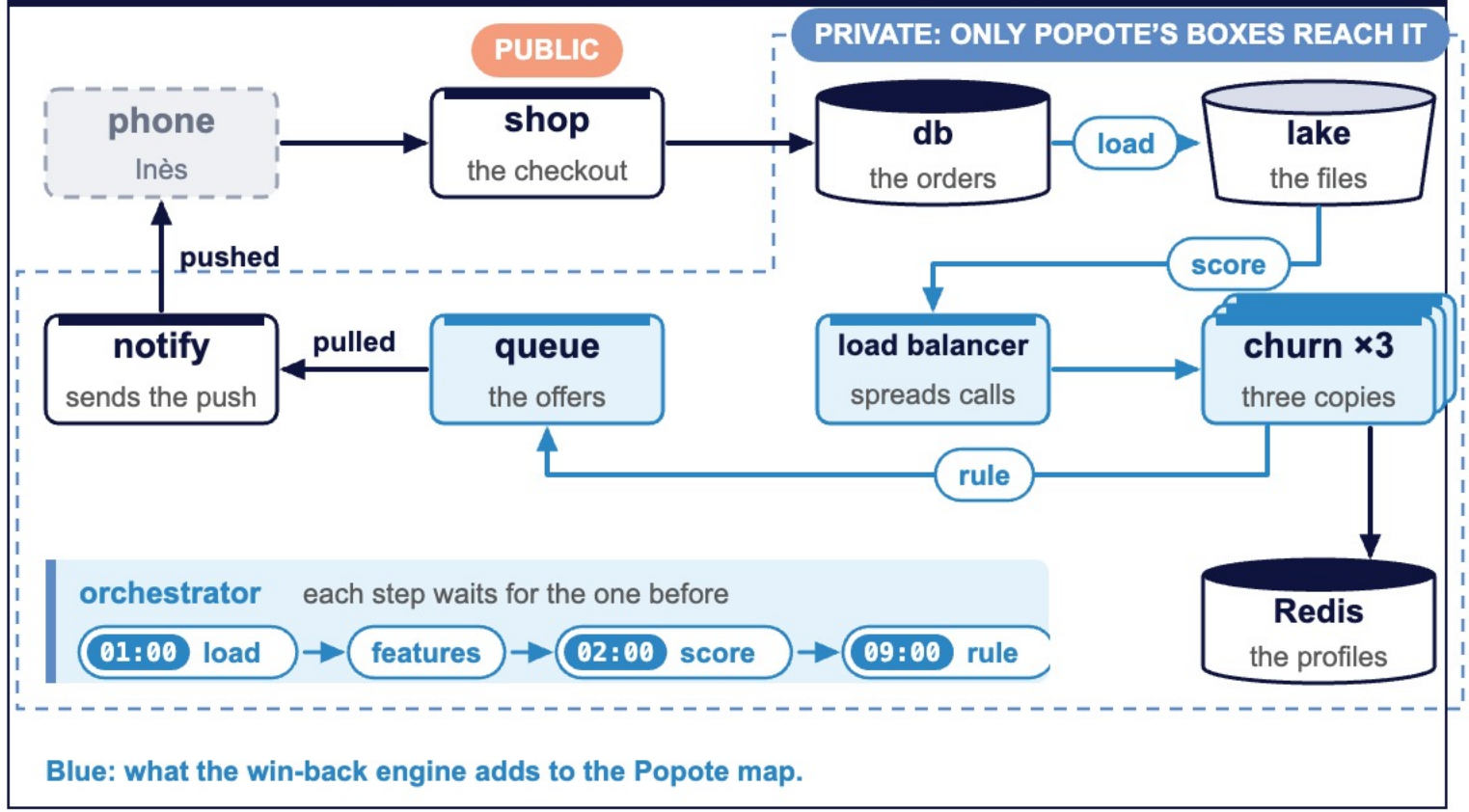
The rest of ARCHITECTURE.md: how the services talk, then every box. Excalidraw recommended.

1 2 3 4 **5 BOXES** 6 7

THE SYSTEM VIEW · HIGH LEVEL



THE DETAILED ARCHITECTURE · LOW LEVEL



Blue: what the win-back engine adds to the Popote map.

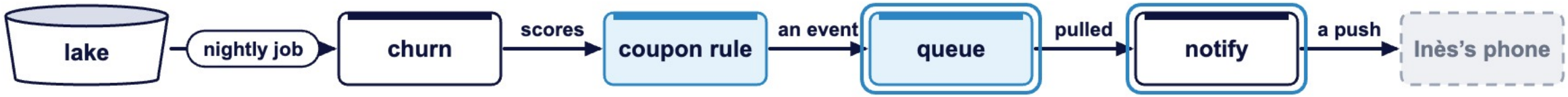
Same product, two zoom levels: the committee reads the first; your operations owner lives in the second.

System view: one arrow, one label (an API call, an event, a nightly job). Detailed: every box, public or private; every arrow, pulled or pushed.

STEP 6: EVERY BOX BUYS SOMETHING AND CAN BREAK SOMETHING

BASICS

The system view, with what each box buys (green) and what it can break (red).



	WHAT IT BUYS	WHAT IT CAN BREAK
the nightly job	+ scores ready by 9:00	− stale: a coupon to someone who ordered at 8:00
the queue	+ the send doesn't block anything	− duplicate: two coupons for one customer
notify	+ she sees it now	− a push after she has already left

ONE DECISION RECORD, IN DECISIONS.md

OPTIONS	Batch at 02:00 · scoring at each visit · both
CHOSEN	Batch at 02:00, and skip anyone who ordered since
WOULD CHANGE IF	Marketing wants offers within the hour

For each box: what you bought, what it can break, and what you decided.

STEP 7: HOW YOU'LL KNOW IT BROKE, AND WHAT YOU DO AT 19:05

BASICS

The heart of OPERATIONS.md for the win-back engine.

1 2 3 4 5 6 **7 RUN IT**

WHAT WE WATCH

coupons sent per day	customers kept: targeted vs held out	nightly job success	queue depth	cost per day

THE THREE LIKELIEST FAILURES

WHAT FAILS	THE ALERT	THE FIRST ACTION
The nightly job didn't run	no scores at 8:00	rerun it, send the coupons late
Two coupons to one customer	duplicate count above 0	check the event id
The model has aged	kept customers drop vs the held-out group	retrain it (session 7)

WHAT AGES OVER SIX MONTHS

the model · the coupon rule · customers who learn to wait for coupons

A product is finished when you know how it breaks, and what you do then.

FIVE CHECKS FOR A GOOD PROJECT

IN WORDS

A good project passes all five. Use them to choose, and to cut.

1. **A real Popote pain:** someone loses money, time or customers today.
2. **One number that moves,** and a way to show it moving on your product (before and after, on the same traffic).
3. **Two or three modules,** not seven.
4. **At least one AI or ML piece,** with a plan for when it's wrong.
5. **A twist of your own:** two groups can pick the same idea, never the same product.

Two or three modules, done well and explained, beat seven half done.

PART 2 OF 3

Idea cards

Examples of modules

Sixteen modules Popote could use, in four tracks: build your business case around them.

1 · ONE IDEA, END TO END | 2 · IDEA CARDS | 3 · HAND IN, ORAL, CALENDAR

SIXTEEN EXAMPLE MODULES: BUILD YOUR BUSINESS CASE AROUND THEM

KICKOFF

Examples, not a list to pick from. Take modules like these, or invent your own.

RETENTION	BASKET	DELIVERY	OPERATIONS AND SUPPORT
Keep the customers we have	**Bigger baskets that fit her life**	**Promises we keep**	**Fewer problems, solved faster**
CARDS A B C D P	CARDS E F G H	CARDS I J K	CARDS L M N O

EVERY CARD HAS THE SAME FIVE LINES		
THE IDEA	the module in one sentence	
THE DECISION	which decision it changes, and who makes it	STEP 1
SUCCESS CRITERIA	the KPIs that would prove it works, and the guardrail that says it isn't doing harm. Make them yours, with targets	STEP 2
BRICKS	what it needs from the course, and the session each brick arrives in	STEP 5
THE HARD QUESTION	the one thing that makes the idea worth a project. Your design must answer it	

A card doesn't tell you what breaks, how it scales or what to watch: finding that is your work (steps 6 and 7).

Take modules like these and build a business case around them. Invent something else if you want: it must pass the five checks.

RETENTION: KEEP THE CUSTOMERS WE HAVE

BASICS

Cards A to D. Each card: the idea, the decision, success criteria, the bricks, the hard question.

THE IDEA	THE DECISION	SUCCESS CRITERIA	BRICKS	THE HARD QUESTION
A The win-back engine Spot who is about to leave and win them back with the right offer, for less than they're worth. **Slides 5 to 11 walk this idea: if you pick it, your twist must go further.**	Who gets a coupon, and how much. Marketing, every morning.	Customers still ordering 30 days later, targeted vs a held-out group that got nothing; coupon spend per customer kept, below what a kept customer is worth. Guardrail: coupons used by customers who would have stayed.	churn (S1), pipeline (S2), notify and A/B (S3), queue (S4).	**How will you know the coupon brought her back, and that she wouldn't have come anyway?**
B The unhappy-customer radar An LLM reads chats and complaints, extracts sentiment and topic, and feeds the churn model.	Which unhappy customers someone contacts today. The support lead.	Churn among complainers contacted vs not contacted; time from complaint to first contact. Guardrail: LLM cost per 1,000 chats, and agreement with chats labelled by hand.	the lake and a pipeline (S2), churn (S1), the chatbot (S5), the model loop (S7).	**Only customers who chat carry the signal: is that fair, and is it worth the LLM bill?**
C The coupon budget optimiser 2,000 € a week: spend it where it changes behaviour.	Where this week's 2,000 € goes. Marketing, weekly.	Retained customers per 1,000 € of coupons, week after week, vs spreading the budget evenly. Guardrail: the budget is never overspent; the same customers don't get every coupon.	churn (S1), warehouse and pipeline (S2), A/B (S3).	**Is the 250th coupon still worth its 8 €, and how would you see it?**
D The late-delivery apology When a promise breaks, apologise in proportion to the delay and the customer's risk.	Which gesture after a late order. Automatic, per order.	Churn after a late order, with vs without the gesture; compensation cost per late order. Guardrail: gestures sent twice, or to orders that weren't late.	churn (S1), route, dashboard and notify (S3), queue (S4).	**How late is late, and what happens when customers learn the rule?**

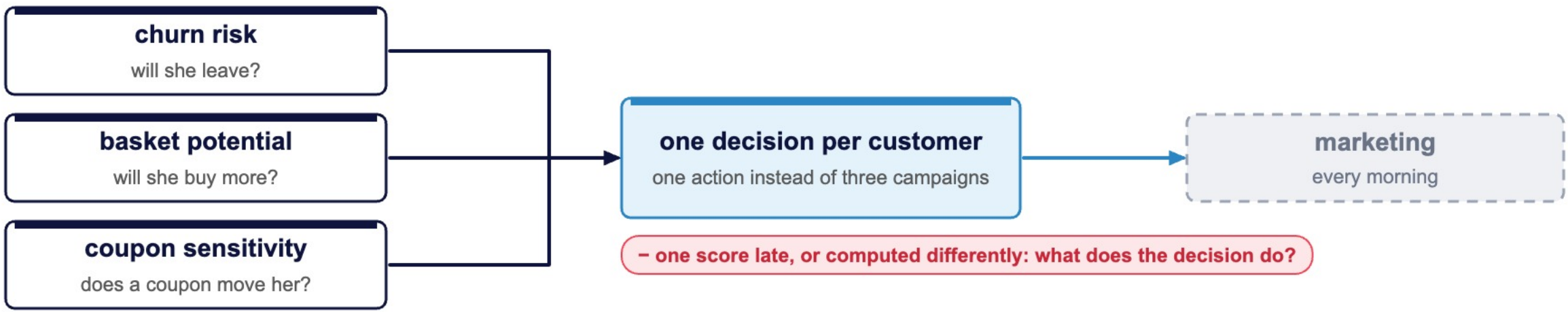
RETENTION: THE CUSTOMER SCORE CARD

BASICS

Card P. Each card: the idea, the decision, success criteria, the bricks, the hard question.

THE IDEA	THE DECISION	SUCCESS CRITERIA	BRICKS	THE HARD QUESTION
P The customer score card Three scores (churn risk, basket potential, coupon sensitivity) combined into one decision per customer for marketing.	One action per customer instead of three campaigns. Marketing, every morning.	Margin per customer with one combined decision vs three separate campaigns; customers hit by two campaigns in a week drops to zero. Guardrail: the decision never uses a score older than a day.	churn (S1), pipeline and warehouse (S2), A/B (S3), the model loop and the feature store idea (S7).	**What does the decision do when one of the three scores is late, or computed differently?**

WHAT CARD P COMBINES: THREE MODELS, ONE SET OF FEATURES, ONE DECISION (SESSION 7)



BASKET: BIGGER BASKETS THAT FIT HER LIFE

Cards E to H. Each card: the idea, the decision, success criteria, the bricks, the hard question.

THE IDEA	THE DECISION	SUCCESS CRITERIA	BRICKS	THE HARD QUESTION
E The meal planner “Dinner for 4, gluten free” becomes a full basket that respects the diet.	Which products go into the basket. The customer, helped by Popote.	Share of suggested baskets ordered; basket size vs a normal order. Guardrail: diet errors in a basket, target zero.	the database (S2), the chatbot and guard (S5), search and meal (S6).	**One forbidden ingredient loses the customer for good: how do you make that close to impossible?**
F The weekly basket autopilot Predict when she’ll reorder and have her usual basket ready.	When to remind her, and with which basket. Automatic, per customer.	Orders per customer per month, reminded vs not reminded; reminder-to-order rate. Guardrail: push opt-outs.	profiles (S1), warehouse and pipeline (S2), notify (S3), search (S6).	**When does a reminder become spam, and how would you notice?**
G The substitution assistant Oat milk is out of stock: offer the closest product she’ll accept, before she cancels.	Which substitute to offer, or swap. The customer, or automatic if she allowed it.	Orders saved when an item is missing; substitutes accepted vs refused. Guardrail: a substitute that breaks her diet, target zero; refunds.	database and ingest (S2), notify (S3), queue (S4), search (S6).	**The closest product in meaning isn’t always the closest in diet or price: who wins?**
H The shopping chat assistant “What can I cook with leeks and potatoes?” The chat answers and fills the basket.	What goes into the basket from a conversation. The customer.	Chats that end in a basket; cost per chat below the extra margin of that basket. Guardrail: wrong products or prices in answers, judged on a sample.	the chatbot, tools, guard (S5), search and retrieval (S6).	**The more customers use it, the more it costs: where does it stop paying?**

DELIVERY: PROMISES WE KEEP

BASICS

Cards I to K. Each card: the idea, the decision, success criteria, the bricks, the hard question.

THE IDEA	THE DECISION	SUCCESS CRITERIA	BRICKS	THE HARD QUESTION
I The honest ETA Promise times you can keep, tuned by hour and store, without scaring customers away.	Which time to promise, per order. Automatic.	Broken promises per store and hour, without losing orders to longer promises (A/B against the fixed rule). Guardrail: conversion at checkout.	warehouse and pipeline (S2), route, dashboard and A/B (S3).	**A shorter promise sells, a broken one loses a customer: where is the line, per store and hour?**
J Proactive delay messaging Tell Inès her order is late before she asks.	When to warn a customer that her order is late. Automatic.	"Where is my order?" chats per 100 orders; ratings of late orders. Guardrail: false alarms; a warning that arrives after the courier.	route, dashboard, notify (S3), queue (S4), the chatbot (S5).	**A false alarm or a late warning: which costs Popote more?**
K The rush-hour guard At 19:00, protect promises: widen them, or pause a store, before customers suffer.	Widen the promises or pause a store. The operations manager, minute by minute at the peak.	Broken promises at the peak; orders lost to pausing, counted against complaints avoided. Guardrail: a store flapping between paused and open.	route, dashboard, notify (S3), queue (S4).	**Pausing loses sure sales to avoid possible complaints: when is it worth it?**

OPERATIONS AND SUPPORT: FEWER PROBLEMS, SOLVED FASTER

BASICS

Cards L to O. Each card: the idea, the decision, success criteria, the bricks, the hard question.

THE IDEA	THE DECISION	SUCCESS CRITERIA	BRICKS	THE HARD QUESTION
L The support copilot A chatbot that solves common questions with real facts and hands over the rest.	Answer the customer, or hand her to a human. Automatic, per chat.	Share of chats solved without a human; cost per chat vs an agent. Guardrail: wrong answers on refunds and policies, judged on a sample, target close to zero.	queue (S4), the chatbot, tools, guard, judge (S5), retrieval (S6).	**One invented refund policy costs more than an agent's chat: how do you keep it honest?**
M Complaint triage Every complaint reaches the right team fast, with nothing lost.	Which team gets which complaint, and by when. Automatic, per complaint.	Time to resolution per team; complaints misrouted. Guardrail: complaints lost between systems, target zero; dead letters left unread.	ingest (S2), dashboard (S3), queue and dead letters (S4), the chatbot (S5).	**A complaint lost between two systems is worse than a slow one: how do you know none is lost?**
N The store manager's morning brief At 7:00, each dark-store manager gets the three things to fix today.	What each store manager fixes first today. The manager, at 7:00.	Stockouts and waste per store, before vs after; manager time saved. Guardrail: every number in the brief matches the warehouse.	warehouse, pipeline and data tests (S2), notify (S3), the chatbot (S5).	**A confident brief with a wrong number: how would anyone catch it?**
O The coupon abuse detector Catch accounts farming welcome coupons without blocking real customers.	Let the order through, check it, or block it. Automatic, at checkout.	Losses from abused coupons avoided. Guardrail: real customers wrongly blocked, and their complaints.	a churn-style model (S1), ingest and database (S2), queue (S4).	**Blocking a real family or a flatshare costs more than one abused coupon: where is the line?**

Pick the idea whose pain and KPI you can name tonight: its hard question is what the committee asks.

PART 3 OF 3

What you hand in

The documents, the oral, the calendar

Who owns what, the 15 minutes in front of the committee, and this week.

1 · ONE IDEA, END TO END

2 · IDEA CARDS

3 · HAND IN, ORAL, CALENDAR

SHORT DOCUMENTS, AND ONE PRODUCT THAT STARTS WITH ONE COMMAND

BASICS

What your repository holds on the day of the oral, and which step each document answers.

PLAN.md

MON 16 NOV, THEN EVERY SESSION

Your business case on one page, sent Mon 16 Nov. Then three lines after every session.

docs/BUSINESS.md

STEPS 1, 2

The pain, the decision, the KPIs and the guardrail, the value line, wrong by half, how you'd prove it.

docs/ARCHITECTURE.md

STEPS 3 TO 5

The functional view, the system view, the detailed architecture. Excalidraw recommended.

docs/DECISIONS.md

STEP 6

One record per trade-off, with your numbers.

docs/OPERATIONS.md

STEP 7

What you watch, the three likeliest failures, what ages and who refreshes it.

docs/CLOUD.md

STEPS 5, 7

Box by box in the cloud (session 7): what runs all day, what scales to zero, the monthly bill. A plan on paper.

packaged_app/

THE PRODUCT

docker compose up starts it. The README: how to run it, three things to try, what doesn't work.

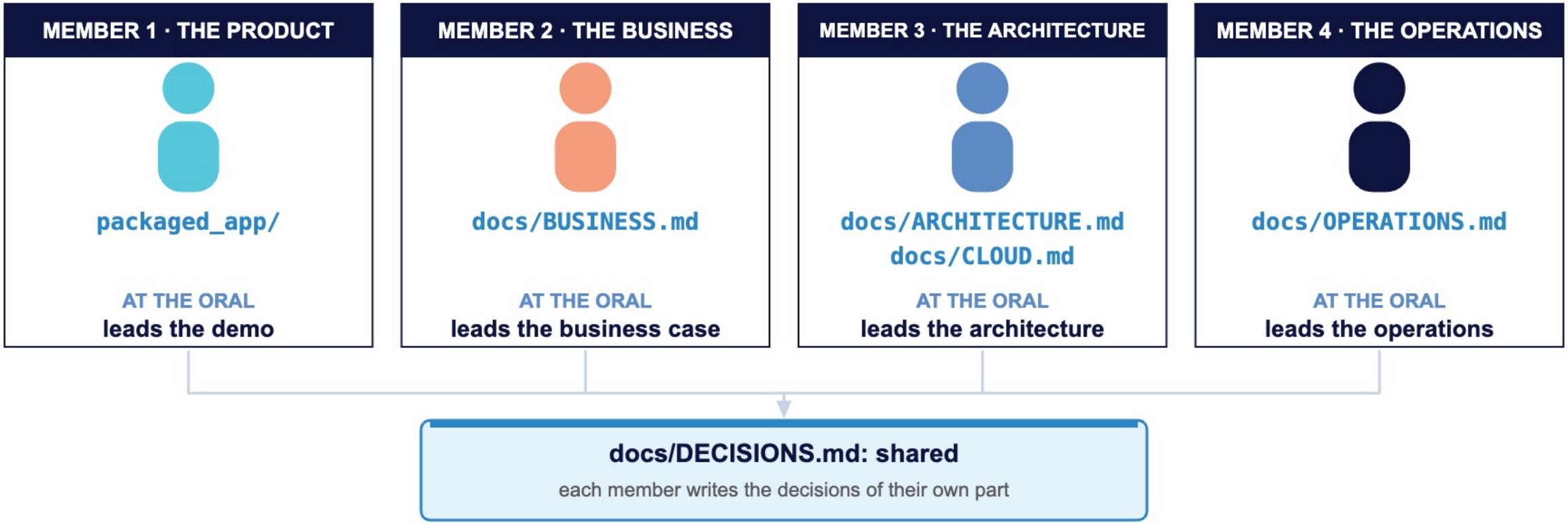
Start from project-template/: every file is there, with its guiding questions.

A page each is fine. The product reuses the lab services: wire them, don't rewrite them.

BASICS

EACH MEMBER OWNS A PART, AND DEFENDS IT ALONE

Four owners. Everyone writes everywhere; the owner answers for it at the oral.



A group of three merges BUSINESS into the ARCHITECTURE owner's part; a group of five splits CLOUD from ARCHITECTURE.

Write the owner's name at the top of each file.

THE ORAL: THREE SLIDES, BUSINESS, ARCHITECTURE, OPERATIONS

BASICS

Fri 27 Nov, 15 minutes per group, in front of Margaux's committee.

0 min	3 min	7 min	11 min	13 min	15 min

3 MIN THE BUSINESS CASE	4 MIN THE ARCHITECTURE	4 MIN THE OPERATIONS	2 MIN THE DEMO	2 MIN QUESTIONS
The pain, the decision, the KPIs and how you'd prove they moved.	The system view, then the detailed architecture; what's public, what it costs in the cloud.	How it stays up at ×100, what you watch, and "it's 19:05 and X fails: walk us through it".	If you built something worth showing (a UI, a dashboard); otherwise how it runs.	One curveball, and one question per member on the part they own.
LED BY **the BUSINESS owner**	LED BY **the ARCHITECTURE owner**	LED BY **the OPERATIONS owner**	LED BY **the product owner**	WHO ANSWERS **picked live**

Three slides, one per part, plus a demo. Bring the laptop with the product already running, and docs/ open.

Send the repository (code and docs/) the same day: the code is run after the oral.

BASICS

ONE WEEK FOR YOUR BUSINESS CASE, THEN A FEW LINES AFTER EVERY SESSION

From tonight to the oral.



MON 9 NOV	MON 16 NOV	S3 TO S7	S8 · MON 23 NOV	FRI 27 NOV
TONIGHT	BEFORE S3	16 TO 20 NOV	FEEDBACK	THE ORAL
Create your group of four.	Send your business case (PLAN.md): the idea, the pain, the KPIs, the bricks you think you'll use.	The last 20 minutes of each session are yours: three lines in PLAN.md, then your project.	Mostly for nailing down your main ideas: architecture, operations, the curveballs, group by group. A mock QCM in the exam's format.	The oral from 09:00, 15 minutes per group; the repository sent the same day. The QCM at 15:00.

You can change your idea until the end of S6: write why in PLAN.md.

One week for the problem; then every session hands you one more brick.

AFTER EACH SESSION: THREE LINES IN YOUR BUSINESS CASE

BASICS

Tonight, then the last 20 minutes of S3 to S7. By the oral, your plan has grown with the course.

AFTER	WORTH BRINGING IN	TO WATCH OUT FOR	WHAT WE CHANGE
every session	what you learnt today that's worth bringing into your product	what could break, cost or age	what you change in your plan, if anything
S3 · the win-back engine	**"a held-out group gives us the A/B test for free"**	**"the push at 09:00 hits 200,000 phones at once at ×100"**	**"send in waves of 10,000, through the queue"**

THE QUESTION EACH SESSION ASKS YOUR PROJECT

S2
Whose pain, what does it cost, which decision changes? Which data does it need, from which store, how fresh?

S3
How does your product reach the customer, and when? What does it promise, and what happens at ×100?

S4
What runs later, with nobody waiting? What fails first, and how does it come back?

S5
Does an LLM earn its cost in your product? What if it's wrong?

S6
Would search by meaning or retrieval help? What goes stale?

S7
What runs where, for how much? How does your model stay good?

S8
What breaks first, and what do you do at 19:05?

Twenty minutes, every session: no blank page the week of the oral.

THIS WEEK, EACH GROUP DOES THREE THINGS

Before Mon 16 Nov, before session 3.

GROUP: .....

- [ ] **1 Create your group of four,**
  and decide who owns which part.

- [ ] **2 Read the idea cards,**
  then take modules like these and shape your business case, or invent your own. Check it against the five checks.

- [ ] **3 Send your business case**
  (project-template/PLAN.md, one page): the idea, the pain, the KPIs, the bricks you think you'll use.

PLAN.md



ONE PAGE

MON 16 NOV, BEFORE SESSION 3

NOT GRADED

No code, no lab, no session you haven't had yet: the problem and the shape of the solution.

NOW: FORM YOUR GROUP, AND WRITE THE PAIN IN TWO LINES

5 minutes, in class, to start: you finish this week.

1. **Four names,**
   and who owns which part.

2. **A track,**
   and two candidate ideas.

3. **Two lines in PLAN.md:**
   “the pain is ... it costs Popote about ... because ...”



5 MINUTES, IN CLASS